Résumé - Che Chen

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Education

University of Michigan *Sep. 2023 - May 2025 (expected)*
M.S. in Robotics

University of Michigan *Sep. 2021 - May 2023*
B.S.E in Computer Engineering, GPA: 3.94/4.00

Shanghai Jiaotong University - UM-SJTU Joint Institute *Sep. 2019 - Aug. 2023 (expected)*
B.S.E in Electrical and Computer Engineering, GPA: 3.26/4.00

Experiences

Robust Controller and System Identification On Robotic Arms *May 2022 - Present*
Research Assistant at ROAHM LAB
- Performed System Identification on Kinova Gen3 to measure the joint friction, damping, and transmission inertia
- Developed a robot control framework that provides both realtime performance and high-level abstraction
- Implemented a robust controller based on interval arithmetic and verified the tracking performance
- Developed gpu algorithms to rapidly calculate edges of 2D zonotopes

Internship at Petoi LLC *May 2022 - Aug. 2022*
- Developed a ROS Interface to control robot powered by OpenCat
- Applied a human pose detection model to map human postures to robot cat [demo]

Realtime and Virtual Driving Simulator [code] *Feb. 2022 - Present*
Multidisciplinary Design Program guided by Prof. Paul Green
- Aims to create a realistic and handy simulation environment for non-programmer researchers
- Developed Joystick Driver on Linux for Carla Simulator
- Developed Wizard program to provide low-cost self-driving car simulation
- Leading software development of the backend team

Localization System Based on UWB (Ultra-Wide Band) Anchors *Feb. 2021 - July 2021*
SJTU Innovative Practical Project (IPP) advised by Prof. Mian Li
- Performed localization task using data from UWB anchors and IMU

Course Projects

Vox-Fusion-Robust [code] [dataset] *Mar. 2023 - Apr. 2023*
Mobile Robotics by Prof. Maani Ghaffari
- Jointly optimizing a per-image embedding for affine transformation so that the SLAM algorithm is robust to global illumination change.
- Created datasets for evaluating both global and local illumination changes

Neural Radiance Field (NeRF) [code] *Apr. 2022*
Deep Learning for Computer Vision by Prof. Justin Johnson
- Learned how to setup PyTorch ML environment from scratch
- Learned key ideas on NeRF such as Volume Rendering methods

Table Tennis Ball Collector for Use in Multi-ball Training [code] *Feb. 2022 - Apr. 2022*
Introduction to Embedded System Design by Prof. Robert Dick
- Implemented a CV algorithm to estimate the 3D position of ping pong balls using a single RGB camera
- Designed a simple but efficient ping pong ball collecter using 3D printing and rubber bands

Robot Localization in PyBullet [code] *Dec. 2021*
Introduction to Algorithmic Robotics by Prof. Dmitry Berenson
- Implemented a path planning algorithm using A* in Pybullet simulation
- Implemented a Kalman Filter and Particle Filter, and analyzed their performance in terms of accuracy

Skills

Programming C++, Python, MATLAB, Julia, HTML/CSS, Bash, LaTeX
Software ROS (1&2), Docker, Git, Emacs, Linux, Solidworks
Knowledge Robot Kinematics & Dynamics, Control System, Robot State Estimation, Computer Vision, Embedded System